

Permissions, umask, PATH

Who can read a file, who can run a script

Mode bits, umask, and PATH

- In a long listing, the nine permission letters are user, group
- Chmod changes those bits; a script needs execute before you can run it with dot-slash
- Umask masks bits off new files so you do not accidentally create world-writable junk
- PATH is a colon-separated search list

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Permissions, umask & PATH

Toggle rwx bits, see octal modes, compute umask results, and debug “command not found” with a PATH lookup lab.

Starter example: mode 644 (~rw-r--r--), umask 022 → new files 644.

Load starter example

Challenges 0 / 22 cleared

Mode 644: Set file mode to 644 (rw-r--r--).

Show hintCheckNextIdle

Mode 644Mode 755Private 600Private dir 700Directory bitd755umask 022 fileumask 022 dirumask 077

which simvwhich lswhich gitwhich vcsnot foundOwner classGroup classOther classexport TOOLS

export LABNo other writeOwner executeWorld readable

Mode bits

☐ Directory

-**I**W-**I**--**I**--

octal 0644chmod 644

rwx

Effective access

For a process matching each class, what can it do?

Owner readyes

Owner writeyes

Permissions lab

Real shell practice

```
wsl - module07-permissions/examples/permissions

# Track A - real Unix
# real shell session (Track A)

$ mkdir -p /tmp/m07_perms && cp -f hello.sh config.txt /tmp/m07_perms/ && cd /tmp/m07_perms && chmod 644 hello.sh
config.txt && ls -l
total 8
-rw-r--r-- 1 yongfu yongfu 145 Jul 20 03:03 config.txt
-rw-r--r-- 1 yongfu yongfu 150 Jul 20 03:03 hello.sh

$ chmod u+x hello.sh

$ ls -l hello.sh
-rwxr--r-- 1 yongfu yongfu 150 Jul 20 03:03 hello.sh

$ ./hello.sh
Hello from hello.sh - permissions example works!

$ chmod 600 config.txt

$ ls -l
total 8
-rw----- 1 yongfu yongfu 145 Jul 20 03:03 config.txt
-rwxr--r-- 1 yongfu yongfu 150 Jul 20 03:03 hello.sh

$ umask
0022

$ umask -S
u=rwx,g=rx,o=rx
```

Real shell — permissions and umask

Real shell practice — try these

```
# ls -l — long listing; read the nine permission letters  
ls -l
```

```
# chmod u+x hello.sh — add execute for your user (symbolic mode)  
chmod u+x hello.sh
```

```
# ls -l hello.sh — confirm the user execute bit appears  
ls -l hello.sh
```

```
# ./hello.sh — run the script in this directory (dot is usually not on  
PATH)  
./hello.sh
```

```
# chmod 600 config.txt — owner read/write only (numeric mode)  
chmod 600 config.txt
```

Pitfalls to watch

- Avoid `chmod seven-seven-seven` unless you truly understand the risk
- Prefer `user-plus-execute` for scripts you own
- Remember: without dot-slash
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, practice `chmod`, `umask`
- When you are ready, take the short quiz, then continue to dotfiles and config homes